

30.6 A 16Mb 166.8TOPS/W Near-Memory Phase-Domain-Computing Ferroelectric NAND Flash for Approximate Nearest Neighbor Search on Edge Devices

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Abstract:
Previous near-memory computing (NMC) or in-memory-computing (IMC) NANDs suffers from limited IO width, large energy-delay-product, and an inability to support diverse vec... [Show More](#)

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 Contents

Approximate nearest neighbor search (ANNS) for sparse, high-dimensional data, such as text embeddings, is widely used in applications like natural language processing, recommendation systems, and vector search [1]–[3]. Deploying ANNS on edge devices enhances real-time responsiveness, enables offline operation, and improves data privacy and security. Near-memory computing (NMC) with NAND flash offers a viable approach to accelerating on-chip ANNS by integrating compute units near the high-density storage. This design reduces latency and energy costs associated with data movement. However, due to high fabrication costs, NMC NAND prototypes are seldom realized or tested in silicon.

Ferroelectric NAND (FeNAND) memory has been under intensive research and development in recent years due to its foundry-compatible process, low write power, high endurance, and scalability [4]– [8]. It therefore provides a promising technology platform for developing next-generation NMC NAND. Yet, several circuit-level challenges persist (Fig. 30.6.1): (1) The small cell pitch and highly parallel readout scheme in NAND impose tight constraints on the area and power budgets for the NMC units; (2) The large bit-line capacitance results in a large energy-delay product (EDP) for readout, hindering performance in frequently accessed scenarios; (3) ANNS workloads for different applications incorporate diverse vector types with various bit-widths, sign formats, and sparsity, whereas prior efforts have focused mainly on binary, unsigned, and dense vector search operations [9]– [11].



To address these challenges, we developed: (1) A near-memory phase-domain-computing (NM-PDC) FeNAND with that integrates highly energy- and area-efficiency PDC units near the BEOL FeNAND array for graph-based ANNS; (2) A charge-discharge interleaving (CDI) sensing scheme with an auto-polarity-switching sense amplifier (APS-SA) that reduces the EDP up to 80.5 % during continuous readout during the search operation; (3) Multi-bit and sign-aware phase-domain computing units (MS-PDCUs) and sparsity-aware input vector processing unit (SA-IVPU) that for highly sparse multi-bit vectors searching. On this basis, we present a 16 Mb NM-PDC FeNAND chip fabricated using a 180 nm logic process with BEOL ferroelectric capacitors. In a search operation, the chip computes the 512 cosine similarities between a 256-dimensional, 4b input vector ($IV_{0\sim 255}[3 : 0]$) and 512 256dimensional, 4b stored vectors ($SV_{0\sim 255,0\sim 511}[3 : 0]$) through highly parallel vector-matrix multiplication (VMM). It achieves an energy-efficiency (EF) of 166.8 TOPS/W for VMM operations, and 12.8 times reduced overall search latency in a typical ANNS task.

[Figure 30.6.2](#) illustrates the overall structure of the NM-PDC FeNAND chip for the graphbased ANNS. It consists of a 16 Mb FeNAND cell array, row decoder and drivers, write and read (W/R) circuitry with 2048 APS-SAs, 2048 MS-PDCUs, 512 similarity distance computing units (SDCUs), one SA-IVPU, and the main control (Ctrl). The FeNAND cell array comprises 128×2048 FeNAND chains (FNCs) with parallel bit-lines (BLs) and source-lines (SLs). Each FNC contains 64 stacked FeNAND cells, which can be addressed by word-lines (WLs), the source-select-gate (SGS), and the drain-select-gate (SGD). The FNC is fabricated by integrating $Hf_xZr_{1-x}O_2$ (HZO)-based metal-ferroelectric-metal (MFM) capacitors in the contact (CT) layer on top of the gate of standard logic transistors. The graph-based ANNS process [\[2\]–\[3\]](#) implemented on the NMC-FeNAND chip is described below. First, we use truncated singular value decomposition (TSVD) to reduce the dimensions (DIMs) of the input query and the dataset to 256. The IVs and SVs are used to construct a graph-based ANNS index, which generates the address bias (ADD_BIAS) to locate different SVs. The IV and ADD_BIAS are then sent to the SA-IVPU for similarity computing. The chip performs VMM between $IV_{0\sim 255}[3 : 0]$ and $SV_{0\sim 255,0\sim 511}[3 : 0]$ in a search operation within 0 to 256 CLK cycles, depending on the IV sparsity, and generates the similarity outputs ($SO_{0\sim 511}[14 : 0]$). The SO is then sent back to the ANNS index to update the ADD_BIAS. This process iterates until no SV with higher similarity can be identified, and then outputs the top-k most similar vectors. The TSVD can reduce the DIM of vectors to reduce data transfer, yet may cause increased recall loss if the DIM becomes excessively small. By truncating the DIM to 256, the data transfer is reduced by 664 times with only a 0.1 % increase in recall loss for the ANNS task using the 20NewsGroup dataset [\[12\]](#).

[Figure 30.6.3](#) illustrates the APS-SA using CDI sensing scheme for continuous read operations. The APS-SA consists of two multiplexers (MUX1 and MUX2) for BL/SL precharge control, a latch-type differential input sense amplifier (SA) to compare the BL voltage (V_{BL}) with the reference voltage (V_{REF}), a DFF to store the previous SA output (SAO), an XOR gate to output SV based on the polarity switching logic, and two switches for read enable (RD_EN) and pre-charge (PRE). The APS-SA is designed to output correct data in the CDI sensing scheme. The polarity (POL) of the APS-SA is set to zero for initialization. A typical read operation is divided into two sub-phases (SP0 and SP1) and activated by RD_EN. In SP0 ($PRE = 0N$), if $POL = 0$, the BL is grounded and the SL is charged to V_{PR} ; if $POL = 1$, the BL is charged to V_{PR} and the SL is grounded. In SP1 ($PRE=OFF$), the SGD of the selected FNC is activated to allow the BL to develop. The BL develops toward V_{PR} if $POL = 0$, or toward GND if $POL = 1$. The SA is then triggered ($SA_EN = 1$) to output the SAO. If the current SAO is the same as the current POL, the POL is kept the same; otherwise, it is switched. By performing an XOR operation between the SAO and POL signals, the APS-SA generates the correct logic value for SV output. The APS-SA with the CDI scheme reduces pre-charge energy by eliminating the need for pre-charge and discharge of the selected BL in each CLK cycle, which is required in the conventional scheme. In addition, the CDI scheme employs a low V_{PR} of 400 mV instead of $V_{PR} = V_{DD}$ in the conventional scheme, further reducing the energy and delay during BL development. Compared to the conventional scheme, the scheme achieves 49.8% lower energy, 59.2% lower delay, and 80.5% lower EDP.

[Figure 30.6.4](#) illustrates the circuit blocks to perform highly efficient VMM in the phasedomain, including the SA-IVPU, MS-PDCUs, and SDCU. The SA-IVPU consists of a sparsity-aware vector aligner (SAVA) and an inverter-based digital-to-time converter with a unit time of τ (τ -DTC). While the SAVA is used to skip over the sparsity in IV and align the IV with the SV, the τ -DTC is used to convert the IV data into the time-domain signal (DTC_OUT). The SAVA is composed of a scan I/O for data input, an 8b counter to count the number of zero-valued entries (N_Z), an 8b WL counter to generate the number of WLs (N_{WL}) for WLs scanning, an adder to calculate the WL address (WADD), and a finite state machine (FSM) for control. To start the vector searching process, N_Z and N_{WL} are reset to 0. $IV_{0\sim 255}[3 : 0]$ is input to SAVA serially. In the i^{th} cycle, if

$IV_i[3:0]$ is zero, N_Z and i are incremented ($N_Z = N_Z + 1, i = i + 1$), and the scan I/O continues scanning in new data until a non-zero value appears or i reaches 256. If $IV_i[3:0]$ is non-zero, WADD is updated as $WADD = ADD_BIAS + N_Z + N_{WL}$ to fetch the corresponding $SV_{i,0 \sim 511}[3:0]$. Then, N_{WL} and i are incremented ($N_{WL} = N_{WL} + 1, i = i + 1$), N_Z is reset to 0, and the next data is scanned in. To perform VMM with $IV[3:0]$ and $SV[3:0]$ in the sign magnitude format, three MS-PDCUs and one SDCU are grouped. Each MS-PDCU consists of one XOR gate for multiplying between the sign bit ($IV[3]$ and $SV[3]$) to generate SIGN signal, one AND gate for multiplying between the unsigned ones ($IV[2:0]$ and $SV[2:0]$), two five-stage voltage-controlled-oscillators (VCOs) for phase accumulation in the positive (SIGN=0) or negative (SIGN=1) directions, a MUX for direction selection, an 8b up and down counter (UDC) and D-type flip-flops (DFFs) to sample the cycles and phase information in the VCO. The SDCU converts the phase data ($PO[4:0]$) to a digital value, multiplies the counter value ($CO[7:0]$) by ten, and then combines these values at their place values using a digital shifter-and-adder (DSA) to produce the $SO[14:0]$. The NM-PDC FeNAND operates in two phases across separate CLK cycles. In the readout phase, the SV is readout from FeNAND according to WADD. In the computing phase, phase-domain VMM is performed in the MS-PDCU. The readout and computing phases operate in a pipelined manner to reduce overall latency. Notably, the unit delay cells of the inverter-based DTCs are designed to be the same as those in the VCO to match their unit time, ensuring robustness against process variation.

[Figure 30.6.5](#) shows the measurement results of the fabricated NM-PDC FeNAND. The FeNAND cells in FNC can be programed and erased (P/E) using fixed-amplitude pulses ($V_{PRG} = 4.0$ V, $V_{ERS} = 3.5$ V) with a duration of 100 ns. A pass voltage (V_{PASS}) of 2.5 V is used for the unselected cells. A clear memory window (MW) > 0.45 V can be maintained after 10^9 P/E cycles. Typical ANNS tasks of searching k similar vectors out of 18 k vectors in the 20NewsGroup dataset are demonstrated on the fabricated chip. Captured functional waveforms show the results for one mismatched and matched sample. A shmoo test confirms that the chip achieves a peak EF of 166.8TOPS/W at 27 MHz and a peak throughput (TP) of 471.0 GOPS at 36 MHz. With the SA-IPVU, the chip achieves a $12.8\times$ reduction in search time and a $2.32\times$ improvement in EF. It achieves recall rates of 92.1% ($k = 100$) and 91.9% ($k = 10$). Compared with previous works on phase-domain computing units and near- or in-memory computing (NMC/IMC) NAND ([Fig. 30.6.6](#)), this work presents a NMC FeNAND chip featuring competitive performance metrics, including cell endurance ($> 10^9$ P/E cycles), storage capacity (16 Mb), and EF (166.8TOPS/W). It achieves a $1.29\times$ improvement in EF [\[16\]](#) and a $2.56\times$ improvement in FoM [\[15\]](#) ($\text{Capacity} \times \text{IVprecision} \times \text{SV-precision} \times \text{Signal Margin} \times \text{EF}$) over prior works. The die micrograph and chip performance summary are shown in [Fig. 30.6.7](#).



Figure 30.6.1:

Challenges for developing NMC NAND for ANNS and the presented (16Mb) HZO-based NM-PDC FeNAND.



Figure 30.6.2:

Structure of the NM-PDC FeNAND, the computing flow for ANNS, and vector dimension optimization.



Figure 30.6.3:

Structure and operation of the APS-SA with CDI-sensing scheme for FeNAND readout.



Figure 30.6.4:

Structure and operation of the NM-PDC units, including SA-IPVU, MSPDCU, and the SDCU.



Figure 30.6.5:
Measurement results of the fabricated NM-PDC FeNAND chip.



Figure 30.6.6:
Comparison with previous works on phase domain MAC units and IMC



Figure 30.6.7:
Die micrograph and chip performance summary.

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